

Example 1

Consider the function

$$P(X = x) = \begin{cases} e^{-x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}.$$

- a) Prove that it is a continuous probability distribution.
- b) Explain why it is incorrect to say that the “probability that $X = 0$ is 1”.

Example 2

In a course, there are 44 male and 8 female students. Suppose they are ranked according to their scores in Quiz 1.

Assume that no two scores are the same and all rankings are equally likely.

Let X denote the highest ranking achieved by a female student. Find $P\{X = 3\}$.

Example 3

A pair of dice are rolled. Assume that the dice is fair. Calculate

- a) the expected value
 - b) the standard deviation
- of the sum of the two dice.

Example 4

In nature, there are some un-explained correlation phenomenon, for example, the EPR paradox (https://en.wikipedia.org/wiki/EPR_paradox). At other times, two random variables are correlated, but in very complicated ways that are difficult to model or analyze, for example, the trends of two stocks.

Consider two fair dice. Assume that the two die are correlated in a mysterious way with a covariance of 1.2. Calculate

- a) the expected value
 - b) the standard deviation
- of the sum of the two dice.

Example 5

There are 151 different first generation Pokémon that can be caught in Hong Kong. Suppose a Pokémon appears at random,

- a) how many types of Pokémon will you expect to catch in 100 successful catches?
- b) how many catches is expected before all Pokémon is collected?

[Hint: See the Coupon Collector Problem

(https://en.wikipedia.org/wiki/Coupon_collector%27s_problem)]

- c) Discuss whether your estimate in b) is reasonable.

Example 6

We have 3 random variables X, Y, Z . Given that X and Z are independent. Simplify $Var(2X - Y + 4Z)$

Example 7

A random process (X, Y) for which X and Y are random variables records the following entries

(1, 2) (2, 4) (3, 6) (4, 0)

Compute the Covariance between the two variables.

Example 8

Find the “expectation of life at birth by sex” from the Hong Kong Census and Statistics Department (<http://www.censtatd.gov.hk/hkstat/hkif/index.jsp>. See under “Health”)

Use the 2021 data below:

Number of births		Crude birth rate (per 1000 population)	Total fertility rate (per 1000 female population) (1) (2)	Number of deaths		Crude death rate (per 1000 population)	Expectation of life at birth		Number of marriages		Crude marriage rate (per 1000 population)	Median age at first marriage		Child dependency ratio (3)	Elderly dependency ratio (4)	Overall dependency ratio (5)
('000)	Year-on-year % change	Rate	Rate	('000)	Year-on-year % change	Rate	(years)		('000)	Year-on-year % change	Rate	(years)		Rate	Rate	Rate
Total	Total	Total	Total	Total	Total	Total	Male	Female	Total	Total	Total	Male	Female	Total	Total	Total
53.7	-5.0	7.2	1 080	47.4	+1.2	6.4	82.3	87.6	49.3	-4.8	6.6	31.5	29.7	159	238	397
52.9	-1.6	7.0	1 064	49.0	+3.3	6.5	82.4	88.1	44.2	-10.3	5.9	31.6	29.9	160	249	409
43.0	-18.6	5.8	883	50.7	+3.5	6.8	83.4	87.7	27.9	-37.0	3.7	31.9	30.4	159	262	421
37.0	-14.1	5.0	772	51.2 p	+1.0 p	6.9 p	83.0 p	87.7 p	26.9	-3.5	3.6	32.2	30.6	157	282	438

You may either use the male figure or the female figure in the question below.

- Provide an estimate of the probability that a person will live to 100 years old using Markov's inequality.
- If the standard deviation of the expectation of life at birth is 5 years, find the probability that a person will live to more than ± 20 years from the expected life at birth.
- Cantelli's inequality is a generalization of Chebyshev's inequality to one sided situations (https://en.wikipedia.org/wiki/Cantelli%27s_inequality). Assume the standard deviation of the expectation of life at birth is 5 years, estimate the probability that a person will live to 100 years old using Cantelli's inequality.
- Comment on your estimates. Also, suggest how you can estimate the expectation of life at birth 2 years later.
- Consider the population pyramid in Hong Kong (e.g. <https://www.populationpyramid.net/china-hong-kong-sar/2021/>)

Go back and forth in time to see how the pyramid is changing.

Also, reference the sex ratio at birth (https://en.wikipedia.org/wiki/Human_sex_ratio)

Suggest a reason why the sex ratio in Hong Kong is dropping.

Example 9

100 samples are taken from an i.i.d. random process. (i.i.d. stands for independent and identically distributed). It means that the samples are taken independently and taking a sample will not affect other samples, for example, the samples are taken with replacement randomly. It also means that the samples are taken from the same (i.e., identical) probability distribution.

What is the lower probability bound if it is required that the sample mean and the true mean differs by less than 2 standard deviations?

Caution: Note that you should not assume that the distribution is Gaussian, as the question does not say so. To test whether a distribution is likely to be Gaussian, you need to conduct statistical tests such as normality test.

Example 10

Suppose a sample of 100 individuals in a large population is taken and the sample mean age is 40 years. Assume the age in the population is i.i.d. and the minimum age is 0 years and the maximum age is 120 years. What is the probability that the true mean lies between 30 to 50 years?

Use the Hoeffding's inequality.

(Hoeffding's Inequality) Let $\theta_1, \dots, \theta_m$ be a sequence of i.i.d. random variables and assume that for all i , $E[\theta_i] = \mu$ and $P[a \leq \theta_i \leq b] = 1$. Then, for any $\epsilon > 0$,

$$P \left[\left| \frac{1}{m} \sum_{i=1}^m \theta_i - \mu \right| > \epsilon \right] \leq 2 \exp(-2m\epsilon^2 / (b - a)^2)$$